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PI Name:	Harrison, Lynn Harrison		
Project Title:	Are Rel and SigH Required for Survival and Biofilm Formation of LSMMG-Grown Mycobacterium marinum?		
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Program/Discipline--Element/Subdiscipline:			
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Key Personnel Changes/Previous PI:	Kara Lind, my postdoc, left the lab in August 2021 to work in industry as an engineer. I have been unable to replace her. I have a rising freshman undergraduate working this summer, and a technician, who has no previous experience but a microbiology degree, will start in July 2022.		
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Performance Goal Text:

Task Description:

Mycobacteria are found in the environment, and some are human pathogens. Mycobacterium tuberculosis, Mycobacterium avium, and Mycobacterium marinum are 3 examples that cause human disease: M. avium and M. tuberculosis result in lung infections, while M. marinum infects the skin. M. avium and M. marinum are waterborne mycobacteria that form biofilms and have been found in biofilms in house showerheads and taps. M. avium was also identified in water from the Mir space station. To date, mycobacteria have not been detected in samples from the International Space Station (ISS). However, deciphering the effects of microgravity on mycobacterial cell signaling and biofilm formation are important to NASA's mission for the following reasons. Firstly, if water on long flight missions is contaminated with mycobacteria not only could there be a problem of biofilm formation in the potable water, but also an increased chance of infection due to astronaut compromised immunity and the increased potential for aerosolized water under microgravity. M. avium can infect humans via aerosolized water. Secondly, we are entering a new era of commercialized spaceflight and people that are not as healthy as astronauts will eventually fly in space. One third of the world's population is believed to be infected with a latent form of M. tuberculosis, which can reactivate under stress conditions and compromised immunity. Thirdly, our transcriptome studies of Low Shear Modeled Microgravity (LSMMG)-exposed M. marinum revealed pathway activation and metabolic changes that are similar to alterations seen in other bacteria exposed to spaceflight and LSMMG, as well as M. tuberculosis inside macrophages. Understanding how mycobacteria adapt to LSMMG could reveal information about general bacterial LSMMG survival mechanisms. It is also possible that growing M. marinum under LSMMG may provide a model to develop and test new drugs against M. tuberculosis. This work could therefore benefit humans on Earth.
The objective of this proposal was to understand the mechanisms used by mycobacteria to adapt to life under LSMMG, and to determine whether their ability to form biofilm is altered under LSMMG. M. marinum was the model organism for this study. It is a good model for the more potent human pathogenic mycobacteria, as it is a Biosafety Level 2 (BSL-2) organism. M. marinum were grown in high aspect ratio vessels attached to the rotary cell culture system, or flaskettes attached to the RPM 2.0 to simulate microgravity. Previously, we were funded by NASA for 1 year to examine how M. marinum altered growth and gene expression under LSMMG. Our previous transcriptome analysis suggests that the Rel and Sigma factor H (SigH) stress response pathways are activated when M. marinum are subjected to LSMMG in regular growth medium. We extended these studies with this proposal. Aim 1 decreased transcription from the Rel, SigH and/ or SigE genes using CRISPR interference to determine how the decrease of these interconnected signaling pathways alters M. marinum growth and antibiotic resistance under LSMMG. Aim 2 examined biofilm/bacteria aggregate formation under LSMMG or simulated microgravity. In flaskettes on the RPM 2.0, the structure of M. marinum biofilm attached to polydimethylsiloxane (PDMS) membrane during growth under simulated microgravity was examined. We also examined the transcriptome response of the bacteria in suspension from the biofilm cultures. This work extends our understanding of mycobacteria subjected to space travel and potentially reveals mechanisms to target mycobacterium in space and on Earth. This work is therefore consistent with the objectives of the NASA Space Biology Program.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Pathogenic mycobacteria such as Mycobacterium avium complex and Mycobacterium marinum have been found in showerheads in homes in the USA. These bacteria form biofilms in the water systems on Earth, and water in showers and hot tubs can result in aerosols containing bacteria. M. avium causes human lung infections and M. marinum predominantly infect human skin, as they grow at 30-33°C. M. marinum is a model organism for waterborne pathogenic mycobacteria. This work is studying the signaling pathways involved in mycobacteria stress responses and trying to understand the responses linked to altered biofilm formation under simulated microgravity. Three genes are being studied to determine if they are important for survival under microgravity. These genes are related to stress and starvation. This work could reveal new targets to develop chemicals against waterborne mycobacteria to reduce biofilm formation in water systems. If new disinfectants/ chemicals can be identified that reduce biofilm formation in household water systems, then this will help reduce non-tuberculosis mycobacteria lung infections on Earth and improve the plumbing of household water systems.

This work uses bacteria called Mycobacterium marinum as a model pathogenic mycobacterium. In the environment, M. marinum is a pathogen of fish and frogs as they grow at temperatures of 28-33°C. M. marinum is therefore less infectious to humans than Mycobacterium tuberculosis and Mycobacterium avium that can result in human lung infections. M. avium and M. marinum form biofilms and have been found in biofilms in house showerheads and taps. M. avium was also identified in water from the Mir space station. Deciphering the effects of microgravity on mycobacterial cell signaling and biofilm formation are important to NASA's mission because of the need to carry water and recycle water during spaceflight. If water on long flight missions is contaminated with mycobacteria, not only could there be a problem of biofilm formation in the potable water, but also an increased chance of infection due to astronaut compromised immunity and the increased potential for aerosolized water under microgravity. M. avium can infect humans via aerosolized water.
Our previous simulated microgravity studies in M. marinum revealed the bacteria grew slower under simulated microgravity compared to normal gravity and uncovered possible signaling pathways required for M. marinum growth under microgravity. Two of the signals that were induced were encoded by the sigH gene and the rel gene. These genes are involved in stress signaling. For example, SigH is known to be necessary to survive heat and oxidative stress. The objective of this proposal was to understand the mechanisms used by mycobacteria to adapt to life under microgravity and to determine whether their ability to form biofilm is altered under microgravity.

Aim 1 examined growth of M. marinum in standard mycobacteria growth medium, which reduces the natural aggregation of the bacteria and allows growth to be more accurately monitored by the ability of the bacteria to form colonies. This medium was used in our previous studies. Cultures were grown in high aspect ratio vessels (HARVs) attached to the rotary cell culture system (RCCS) to simulate microgravity or to expose the culture to normal gravity.

In Aim 1, the importance of SigH, SigE and Rel signaling pathways to growth under simulated microgravity were studied. We used CRISPR interference (CRISPRi) to disrupt expression of SigH, SigE, and/ or Rel. [Ed. Note: CRISPR is an acronym for Clustered Regularly Interspaced Short Palindromic Repeats.] During the first part of the grant, M. marinum strains were produced with the CRISPRi system targeting the specific genes. We optimized the CRISPRi system to down-regulate sigH, sigE, and/ or rel. Studies using oxidative stress or heat shock were used to test the

Task Progress:	efficacy of the CRISPRi system. Decreasing sigH sensitized M. marinum to heat, as expected, and reduced the induction of sigH-regulated genes known to be increased by these stress conditions. We established strains that express decreased levels of sigE or rel, or both sigE and rel or both sigE and sigH. As we previously found, growth under simulated microgravity in the HARVs was reduced compared to normal gravity, and this was not altered by decreasing these signaling pathways. The strain with reduced expression of both sigH and sigE had reduced growth under normal gravity and simulated microgravity. Studies have also been performed to examine survival of M. marinum treated with rifampicin and grown under normal gravity or simulated microgravity. Rifampicin is an antibiotic that is used in combination with other antibiotics to treat mycobacteria infections, including tuberculosis. The rifampicin is dissolved in Dimethylsulfoxide (DMSO) and so the control for zero rifampicin was performed with 0.05% DMSO. Studies were performed in the 10 mL HARVs with exposure to +/- rifampicin and normal gravity or simulated microgravity for 4 days or 24 hrs. Although there was a trend of increased survival to rifampicin under simulated microgravity at 24 hrs, the differences were not significant. However, there was a significant increase in survival of the bacteria to rifampicin when exposed to simulated microgravity for 4 days using the RCCS. A strain with reduced sigH also showed an increase to rifampicin survival under simulated microgravity. Mycobacteria are very “sticky” and aggregate and form biofilms. The regular growth medium contains Tween 80, protein and sugar to reduce biofilm formation. In Aim 2, we did grow the M. marinum in the HARVs using biofilm-promoting medium and overcame technical difficulties where bacteria stuck to the hydrophobic membrane of the HARVs. We determined that aggregate formation and growth was increased due to simulated microgravity. This was unexpected as our studies using the standard growth medium has always found that growth is reduced under simulated microgravity. M. marinum in biofilm-promoting medium were also grown in flaskettes (a small flask attached to a microscope slide). We were able to attach a small piece of hydrophobic polydimethylsiloxane (PDMS) membrane to the slide of the flaskette and sterilize the unit using gamma-radiation. Cultures were initiated in the flaskette, transported overnight to NASA Kennedy Space Center upside down to prevent attachment to the PDMS, and then exposed to simulated microgravity using the RPM 2.0 for 4 days. The normal gravity samples were positioned with the flaskettes the right way up in the incubator so the bacteria could attach to the membrane. We did try cultures of M. marinum expressing red fluorescent protein (RFP) or M. marinum-deficient in rel or sigEsigH. After 4 days, RNA was prepared from the bacteria in suspension and the bacteria attached to the PDMS were fixed with paraformaldehyde and microscopy performed at the Louisiana State University Health Sciences Center (LSUHSC). Attempts were made to examine the biofilm of the M. marinum deficient in rel or sigEsigH by staining for the extracellular matrix in the biofilm. This did not prove successful. We did image the RFP-expressing M. marinum and determined the structure of the biofilm was altered by simulated microgravity. The RNA from RFP-expressing M. marinum was used for transcriptome studies. Surprisingly, sigH and genes regulated by SigH were down-regulated under simulated microgravity. Again, this is very different from our previous studies using standard growth medium. Work has been initiated to study overexpression of SigH and biofilm formation under simulated microgravity using a lab-developed 3D clinostat. This work extends our understanding of mycobacteria subjected to space travel and potentially reveal mechanisms to prevent mycobacteria biofilm formation in space and on Earth.
Bibliography Type:	Description: (Last Updated: 02/27/2023)
Articles in Peer-reviewed Journals	Clary JL, France CS, Lind K, Shi R, Alexander JS, Richards JT, Scott RS, Wang J, Lu X-H, Harrison L. "Development of an inexpensive 3D clinostat and comparison with other microgravity simulators using Mycobacterium marinum." Frontiers in Space Technologies. 2022 Oct 28;3:1032610. https://doi.org/10.3389/frspt.2022.1032610 , Oct-2022